SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50% QUESTIONS: 4

Annexure A

Code of Conduct:

I **K.Reshma(192565062)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Reshma

Industry Problem Task

Scenario: A telecom operator has collected network records containing signal strength, packet loss rate, call drop count, tower load, and weather conditions to build a predictive model for identifying likely network faults. The operator also wants an AI agent to recommend self-healing actions: Reroute Traffic, Restart Node, Increase Bandwidth, or Monitor.

Task 1: Data Preparation & Inductive Learning

1(a) Inductive learning approach, target variable and features

Inductive learning learns a general rule or model from previously observed examples and then uses that model to predict unseen cases. For this telecom problem, historical network records are used as training examples. Each record contains network measurements and a known outcome such as Fault or Normal. A supervised inductive learning model can learn the relationship between the input features and the fault outcome.

Target variable: Network_Fault (1 = Fault, 0 = Normal).

- Signal strength – weak signal can increase communication errors and make faults more likely.
- Packet loss rate – high packet loss directly indicates poor network performance and can be an early fault indicator.
- Call drop count – frequent dropped calls are a strong indicator of network instability.
- Tower load – an overloaded tower can cause congestion and service degradation.
- Weather conditions – severe weather can affect wireless communication and infrastructure.

1(b) Handling class imbalance

The dataset is expected to contain many more Normal records than Fault records. If the model is trained directly on this imbalanced data, it may predict Normal for most cases and still obtain high accuracy while missing important faults.

Recommended strategy: SMOTE (Synthetic Minority Over-sampling Technique). SMOTE creates synthetic minority-class examples from existing fault records. It is suitable here because fault events are rare but important. It gives the model more examples of fault conditions without simply duplicating the same rows.

Class weights can also be used as an additional or alternative method. A higher penalty can be assigned to misclassifying Fault records, which encourages the model to improve recall. Random under-sampling is less preferred when the dataset is not extremely large because useful Normal-operation information may be discarded.

1(c) 70:30 split and preprocessing

Use a stratified 70:30 train-test split: 70% for training and 30% for testing. Stratification is important because it keeps approximately the same Fault/Normal proportion in both sets. A 30% test set gives a sufficiently large independent sample for evaluating a reliability-focused model.

Preprocessing steps:

- Handle missing values using an appropriate method such as median imputation for numeric variables and mode imputation for categorical variables.
- Encode categorical weather conditions using one-hot encoding.
- Normalize numeric variables when using scale-sensitive statistical models such as Logistic Regression.
- Fit preprocessing only on the training data and apply the learned transformation to the test data to avoid data leakage.
- Apply SMOTE only to the training data, never to the test set.

Normalization places numeric variables on comparable scales and can improve optimization for Logistic Regression. One-hot encoding converts categorical values into numeric indicators. Decision Trees generally do not require normalization.

Task 2: Decision Tree for Network Fault Detection

2(a) Decision Tree and first three levels

A Decision Tree recursively divides the training data using feature conditions. The root should be the feature that gives the largest reduction in impurity. Information Gain is commonly used with entropy, while Gini Index measures node impurity.

Illustrative first three levels (actual root and thresholds must be calculated from the supplied dataset):

Level 0: Packet Loss Rate

```
├── Low  → Level 1: Call Drop Count
│         ├── Low  → Normal
│         └── High → Level 2: Signal Strength → Fault/Normal
└── High → Level 1: Tower Load
          ├── Low  → Level 2: Signal Strength → Fault/Normal
          └── High → Fault
```

Justification: packet loss rate is a plausible root because it directly reflects communication quality. However, the correct root must be selected by calculating Information Gain or Gini reduction on the actual training data. For Information Gain, choose the feature with the highest value. For Gini-based splitting, choose the split with the lowest weighted Gini impurity.

2(b) Model evaluation and False Negatives

The assessment file provides no numerical dataset or prediction results, so exact Accuracy, Precision, Recall and F1-Score cannot be calculated from the source alone. The correct formulas are:

- Accuracy = (TP + TN) / (TP + TN + FP + FN)
- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)
- F1-Score = 2 × Precision × Recall / (Precision + Recall)

A False Negative (FN) occurs when an actual network fault is predicted as Normal. In a telecom network, FNs are especially serious because an undetected fault can continue to affect customers. To reduce FNs, use class weights, SMOTE on training data, tune the decision threshold, and optimize for recall or a recall-focused metric. The final threshold should be selected using validation data rather than the test set.

2(c) Post-pruning and deployment choice

Pre-pruning stops tree growth early by setting conditions such as maximum depth, minimum samples per split, or minimum samples per leaf. Post-pruning first grows a larger tree and then removes branches that do not sufficiently improve the objective. Cost-complexity pruning uses a parameter α to balance training fit against tree complexity.

Because the source file contains no dataset or measured results, an exact pre-pruning versus post-pruning accuracy comparison cannot be reported. In practice, compare both models on the same validation/test data using accuracy, recall, F1-score and preferably fault-class recall.

For live network monitoring, a pruned model is generally preferable when it maintains high fault recall while reducing overfitting and unnecessary complexity. The selected model should be the one that performs reliably on unseen network data, not simply the one with the highest training accuracy.

Task 3: Statistical Learning for Risk Stratification

3(a) Logistic Regression / Naive Bayes comparison

Logistic Regression is a suitable statistical learning model because the target is binary: Fault or Normal. It estimates the probability that a record belongs to the Fault class using a weighted combination of the input features.

The assessment asks for Accuracy and AUC-ROC comparison with the Decision Tree. Since the source file contains only the problem statement and no dataset or trained-model output, numerical Accuracy and AUC-ROC values cannot be honestly supplied.

Use the following comparison after training both models:

Model	Accuracy	AUC-ROC	Main strength
Decision Tree	Calculate from test set	Calculate from predicted probabilities	Captures non-linear rules and is easy to interpret
Logistic Regression	Calculate from test set	Calculate from predicted probabilities	Simple probabilistic model and useful for risk scoring

A statistical model is preferable when a stable, interpretable probability or risk score is required, relationships are approximately linear on the transformed features, and consistent probability estimates are more useful than complex rule-based splits.

3(b) Feature importance

For a Decision Tree, feature importance can be obtained from the total impurity reduction contributed by each feature. For Logistic Regression, standardized coefficients can be compared by their absolute magnitude. If the two methods identify similar top predictors, this strengthens confidence that those variables contain important fault-related information.

The likely candidates from the problem description are packet loss rate, signal strength, call drop count and tower load. However, the assessment source does not provide data from which the actual top three predictors can be calculated. Therefore, the exact top-three ranking and whether both models agree must be determined after training on the dataset.

Task 4: Reinforcement Learning for Action Recommendation

4(a) MDP formulation

MDP Component	Definition
States	Represent the current network health, for example S0 = Healthy, S1 = Warning, S2 = Degraded, S3 = Critical.
Actions	Reroute Traffic, Restart Node, Increase Bandwidth, Monitor.
Reward	Positive reward for improving network stability and service quality; small/zero reward for no improvement; negative reward for worsening the condition, unnecessary intervention, or service disruption.
Transition dynamics	After an action, the network moves to another health state according to the observed change in packet loss, signal strength, call drops, tower load and related measurements.
Policy	A mapping from each state to the action with the highest expected long-term return.

This is an MDP because the agent observes a network state, chooses an action, receives a reward, and moves to a new state. The objective is to maximize cumulative future network stability rather than only the immediate reward.

4(b) Q-Learning for at least 5 episodes

Q-Learning updates the value of a state-action pair using the Bellman update:

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha\, [r + \gamma \max Q(s',a') - Q(s,a)]}$$

Example settings: learning rate $\alpha = 0.5$, discount factor $\gamma = 0.9$, and initial Q-values = 0. The following is an illustrative calculation because the assessment source does not provide actual transition/reward data.

Iteration	State	Action	Reward	Next state	Old Q	New Q
1	Degraded	Reroute Traffic	+5	Warning	0	2.50
1	Critical	Restart Node	+8	Degraded	0	4.00
2	Warning	Increase Bandwidth	+4	Healthy	0	2.00

2	Degraded	Reroute Traffic	+5	Warning	2.50	3.75

For example, in iteration 1, if Q(Degraded, Reroute Traffic) = 0, reward = 5, and the maximum Q-value in the next state is 0, then the new Q-value is $0 + 0.5 \times [5 + 0.9(0) - 0] = 2.5$. The agent should be trained for at least five episodes, with each episode containing a sequence of state-action-reward transitions.

Convergence criterion: stop when the maximum absolute change in Q-values between successive training iterations is below a small threshold such as $\varepsilon = 0.001$ for a sufficiently long evaluation window, or when the policy remains unchanged for a predefined number of episodes.

4(c) Final learned policy and ethical considerations

Illustrative policy after learning:

Network State	Recommended Action
Healthy	Monitor
Warning	Increase Bandwidth
Degraded	Reroute Traffic
Critical	Restart Node

This policy is illustrative; the actual final policy must be extracted using argmax_a Q(s,a) after training on real or simulated transition and reward data.

Ethical and operational considerations:

- Service outages: An incorrect autonomous action can worsen an outage or interrupt service. High-risk actions should have safeguards and rollback mechanisms.
- Accountability: The operator must be able to identify which model, state, action and policy version caused an intervention.
- Explainability: The system should log the observed state, selected action, reward and relevant network indicators.
- Human oversight: Critical or high-impact actions can require human approval or an emergency override.
- Safety constraints: The agent should be restricted from actions that could damage infrastructure or create cascading failures.
- Monitoring and auditing: Performance should be continuously monitored for changes in network behavior and model degradation.

Important Note on Numerical Results

The uploaded assessment contains the questions and scenario, but it does not include the underlying telecom dataset, confusion matrix, trained-model results, or Q-Learning transition/reward log. Therefore, exact numerical values for Decision Tree Accuracy/Precision/Recall/F1, Logistic Regression Accuracy/AUC-ROC, actual feature-importance rankings, and a genuinely learned Q-table cannot be derived from the PDF alone. The numerical examples above are explicitly labeled illustrative and should not be presented as measured results.

Source

Prepared from the uploaded document: CO4-AT1-Slot.._260827130832.pdf, SIMATS Engineering, Artificial Intelligence (CSA17), CO4 Assessment Tool 1.